Rayhan Chowdhury

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SUMMARY

Software Engineering student with a Data Science major who enjoys taking a model out of the notebook and putting it in front of real users. I've built and deployed machine learning and deep learning projects end to end, using scikit-learn and PyTorch for the models and FastAPI, Docker and Streamlit to serve them. Along the way I've sorted out the usual deployment headaches, like dependency conflicts. Right now I'm working on deep learning for imbalanced data, starting with credit card fraud detection.

EDUCATION

Daffodil International University March 2023 – Present
B.Sc. in Software Engineering, Data Science Major

PROJECTS

Credit Card Fraud Detection – Deep Learning June 2026 – Present
Work in progress

- Building a deep learning model to flag fraudulent card transactions, with a focus on handling severe class imbalance and balancing precision against recall for real-time screening.

Student Mental Health Prediction
End-to-end ML solution that predicts student mental health scores, deployed as a live web app

- Predicts mental health scores from behavioral, academic, lifestyle and social media factors, covering data analysis, feature engineering and model optimization.
- Tuned a Random Forest model with GridSearchCV and reached a test R^2 of 0.8936.
- Wrapped the model in a FastAPI backend with Pydantic validation and deployed it on Render with an HTML, CSS and JavaScript front end.

  Tech: Python, Pandas, NumPy, scikit-learn, FastAPI, Pydantic, Joblib, Render

  GitHub | Live Demo

Diabetes Risk Prediction – Deep Learning May 2026
PyTorch neural network for diabetes classification, containerized and served as an API

- Built a PyTorch MLP (16-8-1 layers, with dropout) on patient health data. It reached 97.15% test accuracy, 100% precision and 66.51% recall.
- Packaged the model with Docker and exposed it through a FastAPI REST endpoint for real-time predictions.

  GitHub

Customer Churn Prediction API April 2026
Production-style ML REST API for telecom customer churn

- Trained a scikit-learn classification pipeline on the Telco Customer Churn dataset using 19 customer attributes such as tenure, contract type, billing and subscribed services.
- Built a FastAPI service with Pydantic request validation that returns a churn prediction and probability in real time. Containerized with Docker and documented through Swagger UI.

  GitHub

Adult Income Prediction March 2026
Classification pipeline with hyperparameter tuning

- Built a Logistic Regression pipeline tuned with GridSearchCV to predict whether a person earns over $50K from demographic data, and deployed it as an interactive Streamlit app.

  GitHub

Luxury House Price Estimator February 2026
Regression model with a custom dark-themed UI

- Trained a Linear Regression model to estimate house prices from property features, with a custom dark-themed Streamlit interface deployed live.

  GitHub

VOLUNTEERING

Software Quality Assurance and Testing Student Welfare Wing – Deputy Secretary Jan 2026 – Present
Dhaka, Bangladesh. Help run student welfare initiatives and coordinate activities for the Software Testing and QA wing.

SKILLS

ML / Deep Learning: Python, scikit-learn, PyTorch, Pandas, NumPy
Deployment / Tools: FastAPI, Docker, Streamlit, Hugging Face Spaces, Render, Git
Other: C, HTML, CSS, JavaScript

CERTIFICATIONS & ADDITIONAL INFO

Prompt Engineering – Great Learning April 2026
Languages: Bangla (Fluent), English (Intermediate) | **Interests:** Algorithmic trading, ML research, gaming, photography, football